

TIC 165535747

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.249656 d
Epoch (t0) 1570.5882 BTJD
Duration 0.0580 d
Depth 3.499 % (34,986 ppm)
SNR 150.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

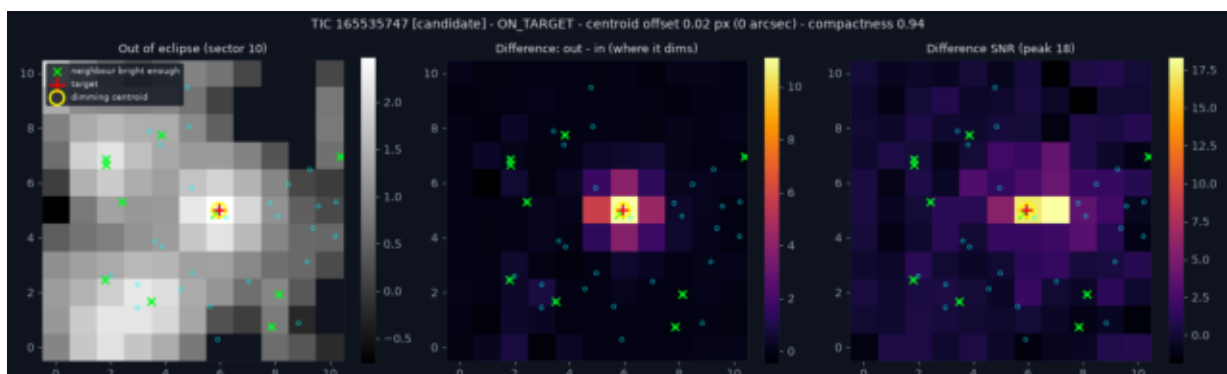
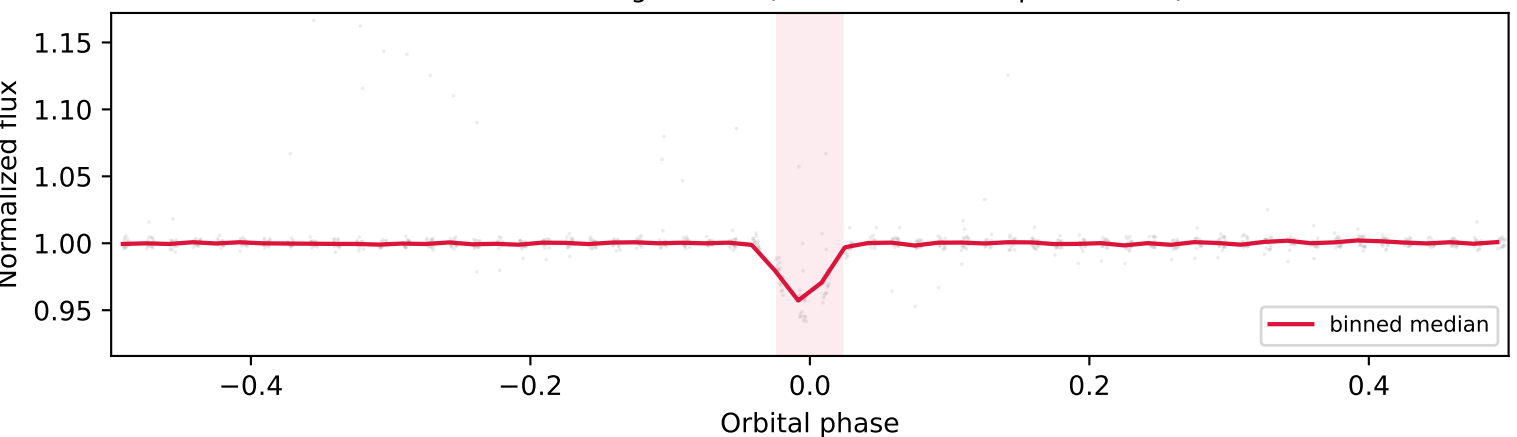
Eclipses observed 13 (gate ≥ 3)
Robust depth ratio 0.90 (gate ≥ 0.50)
Robust depth sigma 66.5 (gate ≥ 7.0)
Odd/even z-score 1.43 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 66.0 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 18.37 (gate ≥ 5.0) Compactness 0.94 (gate ≥ 0.50)
Dimming centroid offset 0.02 px (0 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 165535746
0.22 px away, dTmag +0.25 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.